

MAX ZHU

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Engineering and Mathematics, Minor in Robotics

Expected Graduation: May 2027

- **Concentrations:** Computing Hardware & Emerging Architecture, Systems & Architecture **GPA: 3.91/4.00**
- **Relevant Coursework:** Advanced Computer Architecture, VLSI and Advanced Digital Design, Processor Design, HW/SW Programming, Embedded Systems, Circuit Analysis, Operating Systems, Digital Signal Processing, Data Structures and Algorithms

SKILLS

Hardware: SystemVerilog, UVM, RISC-V, FPGA, Quartus, Vivado, GTKWave, Oscilloscopes, ESP32, UART, I2C, SPI, CAD

Software: C, C++, Python, cocotb, Perl, Java, FreeRTOS, Linux, MATLAB, Django, SQL, HTML, Git

EXPERIENCE

Silicon Jackets

Aug 2024 – Present

Digital Design and Verification Engineer

Atlanta, GA

- Driving RTL development and verification across two tapeouts for a 100MHz RV32I ALU, shifter, and L1 data-cache pipeline
- Authoring micro-architectural specifications and block diagrams, translating system requirements into synthesizable SystemVerilog
- Established a robust verification infrastructure using a UVM env (agents/scoreboard) and an ISS reference to achieve 100% functional and $\geq 95\%$ code coverage, revealing 10 edge cases through automated Python/CI regression pipelines

FUSMobile

Apr 2025 – Aug 2025

Embedded and QA Engineer

Alpharetta, GA

- Engineered a bidirectional USB serial protocol between an Arduino Due and an imaging workstation to enable closed-loop automatic self-alignment, parsing live X-ray inputs from the Neurolyzer to drive real-time spatial orientation
- Built an automated medical report generation system in C++, using MFC for metadata extraction, OpenCV for image filtering/processing, and libHaru for annotation, reducing manual reporting time by 90%
- Re-engineered the Neurolyzer, increasing portability by 20% and interactivity, for hands-on conference demonstrations

CRNCH Rogues Gallery

Jan 2025 – May 2025

Reconfigurable Researcher

Atlanta, GA

- Designed and verified a SystemVerilog RTL implementation of the XChaCha20 encryption algorithm on a PYNQ-Z2 FPGA achieving 10 $\times$ higher efficiency when encrypting and decrypting compared to a C-based CPU implementation
- Automated FPGA build and simulation environments using Vivado, Docker, and PYNQ, for cryptographic workloads

MARGA Lab

Aug 2024 – May 2025

Wireless Digital Twin Researcher

Atlanta, GA

- Created Python scripts with Sionna to optimize transmitter-receiver placement by over 60% in over 50 unique environments
- Analyzed and annotated 30+ research papers on 5G, Wi-Fi 6, ray tracing, and UAVs to establish baseline models
- Published and presented a paper on varying geometric simplifications applied to Blender environments for WDTs, reducing simulation rendering time by 50% while preserving over 95% signal fidelity compared to full-scale models

PROJECTS

UVM Memory Verification Environment | SystemVerilog, UVM, AXI

- Developed a reusable UVM environment to verify a memory subsystem with cache and AXI-connected SRAM
- Implemented driver, monitor, scoreboard, and predictor components, enabling constrained and random coverage-driven testing
- Created sequences exercising cache hits, misses, evictions, conflicts, and writebacks across different AXI burst patterns
- Wrote SVA assertions for AXI protocol compliance and integrated functional coverage for eviction types and pipeline events

RISC-V Pipelined Processor | SystemVerilog, RISC-V, GTKWave

- Designed a 5-stage RISC-V processor extended with branch prediction, out-of-order execution, and external ALU
- Integrated a branch predictor (BHR, PHT, BTB), reducing pipeline flushes and improving CPI by 10% on a Towers of Hanoi benchmark
- Developed testbenches to validate structural, data, and control hazards, tracing execution and debugging violations via GTKWave

EXTRACURRICULARS

Leadership: GT IEEE (Director of Corporate Affairs), Hive (Master Peer Instructor), ECE Student Ambassadors

Clubs: Silicon Jackets (Digital Designer), Chinese Student Association (Dynasty Head), Trading@GT (Prediction Markets), Phi Delta Theta

Interests: Motorcycles, Weightlifting, Algorithmic Trading, Cars, Snowboarding, Keyboards, Photography